

HADIA ALVI

AI Engineer

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PROFILE SUMMARY

AI Engineer and Full-Stack Developer with experience in Machine Learning, Computer Vision, and scalable web applications. Skilled in Python, TensorFlow, PyTorch, OpenCV, Next.js, NestJS, MongoDB, and PostgreSQL. Passionate about building AI-powered solutions and automation systems that solve real-world challenges.

WORK EXPERIENCE

Remote Internship <i>Zimo Group</i> Domain: Full Stack Development	07/2025 – 01/2026
Remote Internship <i>Developers Hub Corporation</i> Domain: AI/ML	03/2025 – 05/2025
Remote Internship <i>Zidio Development</i> Domain: Full Stack Development	01/2025 – 02/2025
Trainer <i>Cyber Security Society</i>	
Freelancing	

EDUCATION

Bachelor of Science in Artificial Intelligence NATIONAL TEXTILE UNIVERSITY • CGPA:3.1	2022 – 2026
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SKILLS

Programming Languages Python, JavaScript, C++	Automation n8n, GoHighLevel
Kali Linux/Parrot OS	Machine Learning TensorFlow, PyTorch, Scikit-learn
Deep Learning CNNs, RNNs, Transformers	Data Science Pandas, NumPy, Matplotlib, Seaborn
Computer Vision OpenCV, Mediapipe	Web Development React.js, Node.js, Express.js, MongoDB
Cloud & Deployment AWS, Firebase, Vercel	Tools & Platforms Jupyter Notebook, VS Code, Git, Docker
Shopify	Wordpress
Mobile Application Development Flutter	

PROJECTS

Outreach & Follow-Up Automation System (n8n) • Designed and developed an AI-powered automation system using n8n to streamline outreach campaigns and follow-up processes for lead management.

- Engineered automated workflows for email, WhatsApp, and CRM integrations to manage customer interactions and engagement tracking efficiently.
- Implemented API-based integrations with external tools to automate data syncing, response tracking, and scheduled follow-ups.
- Built intelligent workflow logic with conditional triggers, reducing manual effort and improving conversion efficiency.

Restaurant Management App(Flutter App)

Designed and developed a cross-platform restaurant management application using Flutter for efficient order handling, menu management, and billing operations.

- Engineered a real-time system for managing customer orders, kitchen workflow, and table assignments with smooth state management.
- Implemented backend integration for data storage, order tracking, and transaction management to ensure operational efficiency.
- Built a responsive and intuitive UI to enhance user experience for staff, customers, and restaurant administrators.

Sign Language Detector (AI/ML)

- Designed and developed a Python-based sign language detection system using computer vision techniques to recognize and interpret hand gestures.
- Engineered image processing and feature extraction pipelines using OpenCV to detect hand movements and key gesture patterns.
- Implemented machine learning models to classify sign language gestures into corresponding alphabets or commands with real-time prediction capability.
- Built a user-friendly interface for live gesture recognition, enabling improved accessibility and communication support for hearing-impaired individuals.

Automated Solar Panel Cleaning System (IoT)

- Designed and developed an IoT and ML-based automated solar panel cleaning system to improve energy efficiency and reduce manual maintenance efforts.
- Engineered sensor-based automation using IoT devices to detect dust levels and trigger cleaning mechanisms autonomously.
- Implemented machine learning models to analyze environmental conditions and optimize cleaning schedules for maximum solar output efficiency.
- Integrated hardware and software components to enable real-time monitoring, automation control, and performance optimization of solar panels.

FYP

Vision Bot – AI-Powered Robotic Surveillance System (FYP)

- Designed and developed an AI-powered autonomous surveillance robot using Computer Vision, IoT, Flutter, Firebase, and TensorFlow Lite to provide intelligent real-time security monitoring and threat detection.
- Engineered a mobile robotic platform integrated with DC motors, ultrasonic sensors, GPS tracking, and microcontroller-based navigation, enabling autonomous patrolling, obstacle avoidance, and location-aware surveillance.
- Implemented real-time AI inference using optimized TensorFlow Lite and Computer Vision models on a smartphone-mounted camera to detect smoke, unknown individuals, facial identities, and crowd activity directly on-device with low latency.
- Built a cloud-connected surveillance architecture using Firebase Firestore, Storage, and Cloud Messaging to store event logs, GPS coordinates, detection images, and deliver instant security alerts to authorized users.
- Developed Flutter-based mobile applications for both AI detection processing and end-user monitoring, enabling live alert notifications, event history tracking, robot location monitoring, and remote surveillance management.
- Integrated end-to-end communication between robotic hardware, AI modules, mobile applications, and cloud services, creating a scalable and cost-effective surveillance solution for educational institutions, residential communities, and security-sensitive environments.
- Optimized system performance through edge computing, on-device AI processing, and intelligent event-based alerting, reducing dependency on expensive surveillance infrastructure while improving threat detection accuracy and response times.